

BHUVANAMITRA S

Geospatial Manager | Remote Sensing & GIS | Agriculture, Infrastructure & Aviation | Project Management
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PROFESSIONAL SUMMARY

Geospatial Manager and Remote Sensing Specialist with 6+ years of hands-on experience delivering end-to-end GIS solutions, satellite data analytics, and spatial intelligence products across agriculture, environmental monitoring, and aviation domains. Proven ability to build and lead cross-functional teams, manage full project lifecycles using Agile methodologies, and translate complex spatial data into actionable client deliverables. Experienced in QGIS, Python-assisted automation, and multi-spectral satellite analysis. Adept at stakeholder coordination, sprint planning, delivery tracking, and documentation using Jira and Confluence. Track record of driving GIS product development, quality assurance, and on-time delivery across multidisciplinary programs.

CORE COMPETENCIES

GIS & Remote Sensing: ArcGIS Pro/Online, QGIS, ERDAS Imagine, Google Earth Engine, GDAL	Project Management: Agile/Scrum, Jira, Confluence, sprint planning, stakeholder coordination
Satellite Data Analytics: Multi-spectral, time-series, crop classification, LULC mapping	Environmental GIS: LULC change detection, EIA support, environmental compliance reporting
Python Scripting & Automation: AI-assisted workflows, raster/vector processing, GDAL	CAD & Design: AutoCAD Civil 3D, Adobe Photoshop, cartographic layout
Geospatial Dashboards & Reporting: Tableau, ArcGIS Dashboards, thematic mapping	Multi-Criteria Analysis: AHP-based site suitability, susceptibility zonation mapping
Aeronautical Cartography: OLS, AMDB, eTOD, ICAO Annex 4/14/15, Electronic Charting	Field Data Collection: Survey design, ground-truth surveys, GPS/DGPS, QGIS field tools
Hydrological Modelling: Streamflow forecasting, CHIRPS validation, watershed analysis	Infrastructure GIS: Transmission line routing, drainage design, RoW & land acquisition analysis

PROFESSIONAL EXPERIENCE

Geospatial Manager | [SatSure Analytics India Pvt. Ltd., Bengaluru](#) | Jan 2023 – Present | 3 years 2 months

Leading geospatial product development, agricultural intelligence, and environmental monitoring initiatives — managing cross-functional teams and delivering data-driven solutions for diverse clients.

- Managed end-to-end project execution lifecycle including requirement gathering, sprint planning, stakeholder communication, risk tracking, and on-time delivery across multiple geospatial programs using Agile methodologies.
- Utilized Jira for project and sprint management, issue tracking, and workflow automation; maintained technical documentation, SOPs, and delivery records in Confluence for cross-team visibility.
- Spearheaded crop classification workflows using multi-spectral, multi-temporal satellite data across diverse agro-climatic zones, enabling precision agriculture insights for agri-finance and agri-insurance clients.
- Led environmental monitoring projects — assessing Land Use Land Cover (LULC) changes to support policy-level decision-making and environmental compliance reporting.
- Built client-facing geospatial dashboards, thematic maps, and analytical reports integrating satellite data, vector layers, and tabular statistics using QGIS and Tableau.
- Developed full-stack geospatial data solutions in SatSure SKIES — including base data preparation and route optimisation for transmission line planning using high-resolution satellite imagery.
- Led product development for GIS-based aviation data tools including automated Obstacle Limitation Surfaces (OLS), Aerodrome Mapping Databases (AMDB), and Electronic Charting modules compliant with ICAO Annexes 4, 14 & 15 — deployed across Indian and international airports.
- Managed and mentored a team of engineers, data analysts, and product managers — driving delivery timelines, quality assurance, and cross-functional collaboration.

- Developed an open-source QGIS plugin (GeoBearing-Distance) for on-the-fly geometry generation using ellipsoidal bearing and distance; published on GitHub.

Team Lead – Remote Sensing Operations | [SatSure Analytics India Pvt. Ltd., Bengaluru](#) | Nov 2021 – Dec 2022 | 1 year 1 month

- Led remote sensing operations team across multiple agriculture and environmental projects; coordinated workload allocation, QA reviews, and timely client delivery.
- Supervised LULC mapping and multi-spectral crop classification using time-series satellite data across diverse geographies; enforced quality thresholds and metadata standards.
- Conducted internal knowledge-transfer sessions and trained junior analysts on GIS tools, satellite data processing, and delivery workflows.

Data Analyst – GIS | [SatSure Analytics India Pvt. Ltd., Bengaluru](#) | Jan 2021 – Oct 2021 | 9 months

- Performed advanced geospatial analysis for environmental impact assessment and LULC projects; produced client maps, spatial statistics, and analytical reports.
- Built geospatial visualisations and dashboards using QGIS and Tableau integrated with satellite-derived datasets.

Junior Data Analyst – GIS | [SatSure Analytics India Pvt. Ltd., Bengaluru](#) | Jan 2020 – Dec 2020 | 1 year

- Led LULC classification project around Desert National Park, Rajasthan — conducted stratified ground-truth surveys and achieved 85% classification accuracy.
- Produced detailed spatial maps and statistical analysis reports for client and stakeholder submission.

Project Trainee – GIS | [SatSure Analytics India Pvt. Ltd., Bengaluru](#) | Sep 2019 – Dec 2019 | 4 months

- Performed unsupervised crop classification using time-series multi-spectral satellite data; extracted statistics and built storyboards for crop studies.
- Evaluated and validated open-source satellite rainfall products (CHIRPS) against ground station records; recommended product was adopted in production, replacing a commercial data source and saving ₹25 Lakhs annually.

KEY PROJECTS

- Railway Crossing Elimination Decision Intelligence Tool — Contributed to conceptualisation and solutioning of a geospatial decision intelligence platform for eliminating manned railway crossings: cost-effective alignment identification, RoW estimation, land acquisition analysis (2013 LARR Act), cost estimation, and scenario-based comparative analysis.
- Crop Classification & Agriculture Intelligence — Multi-temporal, multi-spectral satellite data analysis for crop-type mapping across agro-climatic zones; delivered for agri-finance and agri-insurance use cases.
- LULC Change Detection & Environmental Monitoring — Land cover change assessment for policy and EIA compliance; Desert National Park, Rajasthan (85% accuracy, ground-truth validated).
- GIS-based Aviation Data Products — OLS, AMDB, eTOD, and Electronic Charting for Indian and international airports; SID protection areas per ICAO Doc 8168.
- Landslide Susceptibility Mapping — AHP and satellite data-based multi-criteria zonation mapping for disaster risk assessment.
- Transmission Line Route Optimisation — GIS-driven spatial analysis and base data preparation for power infrastructure planning using high-resolution satellite imagery.
- Satellite Rainfall Data Evaluation (CHIRPS) — Validation of open-source rainfall products against ground station data for hydrological forecasting; outcome adopted in production saving ₹25 Lakhs.
- GeoBearing-Distance QGIS Plugin — Open-source plugin for geometry generation using ellipsoidal bearing and distance; published on GitHub.
- Hydrographic Surveying — Sandbar identification in CRZ regions of three rivers in Udupi District, Karnataka.

EDUCATION

M.Tech. (Research) – Remote Sensing & GIS | [National Institute of Technology Karnataka \(NITK\), Surathkal](#) | Aug 2017 – Sep 2019

CGPA: 7.92 | Thesis: Evaluating the Potential of CHIRPS Rainfall Data in Streamflow Forecasting Using Soft-Computing Techniques | Supervisor: Prof. Paresh Chandra Deka

PUBLICATION

Sulugodu, B. and Deka, P.C. (2019). Evaluating the performance of CHIRPS Rainfall Data in Streamflow Forecasting. Water Resources Management. DOI: 10.1007/s11269-019-02340-6

TECHNICAL SKILLS

GIS & Remote Sensing: QGIS, ArcMap/ArcGIS Pro, ERDAS Imagine, Google Earth Engine, GDAL

Programming & Scripting: Python (AI-assisted automation), SQL

Project & Collaboration Tools: Jira, Confluence, MS Office (Excel, Word, PowerPoint), Tableau

CAD & Design: AutoCAD Civil 3D, AutoCAD Revit Architecture, Adobe Photoshop

Standards: ICAO Annexes 4, 14 & 15 | EUROCONTROL ED-99D / ED-119C | DGCA India CAR

ACADEMIC OUTREACH & TRAINING

Conducted 5 workshops and guest lectures on GIS, Remote Sensing, and Geospatial Data Analytics for undergraduate and postgraduate students across institutions including NIE Mysuru (IEEE GRSS), Bharati Vidyapeeth, and SMVITM — covering QGIS, satellite data interpretation, field data collection, and data-driven decision making.

AWARDS & RECOGNITION

- ISTE Best Student Award — ISTE Karnataka Section
- Shri Madhwa Vadiraja Prashasthi Best Student Award 2017 — SMVITM, Bantakal

LANGUAGES

English (Professional Proficiency) | Kannada (Native)